$$\vec{r} = \vec{a} + s(\vec{b} - \vec{a}) + t\vec{c} \quad \text{or} \quad \vec{r} = (1-s)\vec{a} + s\vec{b} + t\vec{c} \qquad \ldots (1)$$

where $s, t \in \mathbb{R}, (\vec{b} - \vec{a})$ and $\vec{c}$ are not parallel vectors.

(b) Non-parametric form of vector equation

Equation (1) can be written equivalently in non-parametric vector form as

$$(\vec{r} - \vec{a}) \cdot ((\vec{b} - \vec{a}) \times \vec{c}) = 0 \qquad \ldots (2)$$

where $(\vec{b} - \vec{a})$ and $\vec{c}$ are not parallel vectors.

(c) Cartesian form of equation

If $\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$, $\vec{b} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$, $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k} \neq \vec{0}$ and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then equation (2) is equivalent to

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

This is the required Cartesian equation of the plane.

Example 6.43

Find the non-parametric form of vector equation, and Cartesian equation of the plane passing through the point $(0,1,-5)$ and parallel to the straight lines $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + s(2\hat{i} + 3\hat{j} + 6\hat{k})$ and $\vec{r} = (\hat{i} - 3\hat{j} + 5\hat{k}) + t(\hat{i} + \hat{j} - \hat{k})$.

Solution

We observe that the required plane is parallel to the vectors $\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$, $\vec{c} = \hat{i} + \hat{j} - \hat{k}$ and passing through the point $(0,1,-5)$ with position vector $\vec{a}$. We observe that $\vec{b}$ is not parallel to $\vec{c}$. Then the vector equation of the plane in non-parametric form is given by $(\vec{r} - \vec{a}) \cdot (\vec{b} \times \vec{c}) = 0$. …(1)

Substituting $\vec{a} = \hat{j} - 5\hat{k}$ and $\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 6 \\ 1 & 1 & -1 \end{vmatrix} = -9\hat{i} + 8\hat{j} - \hat{k}$ in equation (1), we get

$(\vec{r} - (\hat{j} - 5\hat{k})) \cdot (-9\hat{i} + 8\hat{j} - \hat{k}) = 0$, which implies that

$$\vec{r} \cdot (-9\hat{i} + 8\hat{j} - \hat{k}) = 13.$$

If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is the position vector of an arbitrary point on the plane, then from the above equation, we get the Cartesian equation of the plane as $-9x + 8y - z = 13$ or $9x - 8y + z + 13 = 0$.

Example 6.44

Find the vector parametric, vector non-parametric and Cartesian form of the equation of the plane passing through the points $(-1,2,0)$, $(2,2-1)$ and parallel to the straight line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$.

Solution

The required plane is parallel to the given line and so it is parallel to the vector $\vec{c} = \hat{i} + \hat{j} - \hat{k}$ and the plane passes through the points $\vec{a} = -\hat{i} + 2\hat{j}$, $\vec{b} = 2\hat{i} + 2\hat{j} - \hat{k}$.

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